Summary of UW-Madison GI Tract Image Segmentation Competition from He

A1 Team Background

Competition Name: UW-Madison GI Tract Image Segmentation
Team Name: He
Private Leaderboard Score: 0.88255
Private Leaderboard Place: 3rd

Name	Location	Email
He	Beijing, China	XXXXXXXX@foxmail.com

A2 Academic and professional background

He works as a computer vision engineer in Beijing. And he graduated from XXX University as a master at 2021, and during the period from 2019 to 2020, participated in thesemantic segmentation competition on Kaggle many times, and has rich experience insemantic segmentation.

Related experience and time used

A lot of relevant experience, such as loss function, model structure, etc., are from solutions in previous competitions. I learned a lot about computer vision from kaggle, which gave me a good understanding of medical image processing.
The purpose of this competition is to detect large bowel, small bowel and stomach in MRI images through computer vision technology. The purpose of participating in this competition is to reduce the burden on doctors through computer vision technology and better help patients. Many previous computer vision competitions has given me a wealth of experience in image processing in medical image.
It costs about three months for me to initiate, build and train models and summary the competition. I spend about two hours a day from Monday through Friday and about 8 to 10 hours a day on weekends.

A3,A4 Model summary

Brief summary for detection part

The main purpose of the detection part is to remove invalid edge images and keep only the body parts for segmentation. My solution in detection part is mainly based on EfficientDet-D0, and traing image size is 256, and epoch is 10, all other parameters use default parameters. This is a very easy task for the model to learn. I generated some training samples using image preprocessing and relabeled the bad bounding box.

Brief summary for classification part

My solution in classification is mainly based on CNN models. I mainly use unet with a branch of classification, and used 4 different backbone of the unet[1], which are efficinetnet-b7-ns[2], efficinetnet-b7[3], efficientnet-v2-l[4] and efficientnet-v2-m. I trained and evaluated on all image data provided by Kaggle and UW-Madison GI Tract Image Segmentation websites. To train each model, I used a cycle learning rate[5] with warm up was set to schedule. I use binary cross entropy loss for all segmentation branch and classification branch. In each model, I use 5 fold cross validation and data split by case, Exponential moving average (EMA) is used during training, and use stochastic weight averaging (SWA) [6] to ensemble each fold's different snapshots. Independent models were finally ensembled through averge from each model. The public tools involved are pytorch, numpy, pandas, opencv. It takes 6 ~ 8 hours to train each single fold model, and the full training process will take 4 ~ 5 days, using a A5000 GPU. Below are the training parameters for the different models:

models(unet)	size/slice	epoch	init lr	optimizer
efficientnet-b7	320/5	35	5e-4	adamW
efficientnet-v2-l	320/3	35	5e-4	adamW
efficientnet-v2-l	320/5	35	5e-4	adamW
efficientnet-v2-m	352/5	35	5e-4	adamW
efficientnet-b7-ns	320/5	35	5e-4	adamW

Brief summary for segmentation part

My solution in segmentation is similar to the classification part. Two different backbone(efficientnet-v2-l and efficientnet-v2-m) of the unet are used, efficientnet-v2-l(v1) and efficientnet-v2-l(v2) use a different pretrained model (320 size classification model and 384 size segmentation model) . I trained and evaluated on positive sample data provided by Kaggle. I use combining loss(bce: 0.5, dice: 0.5, lovasz: 1) function for segmentation branch and use binary cross entropy loss for classification branch, and all segmentation models use classification models or segmentation models(small scale) as pretrained models to improve training speed. Other settings are the same as in the image classification part. It takes 6 ~ 8 hours to train each model, and the full training process will take 3 ~ 4 days, using a A5000 GPU. Below are the training parameters for the different models:

models(unet)	size/slice	epoch	init lr	optimizer
efficientnet-v2-l	384/5	35	3e-4	adamW
efficientnet-v2-l(v1)	416/5	35	3e-4	adamW
efficientnet-v2-l(v2)	416/5	35	3e-4	adamW

models(unet)	size/slice	epoch	init lr	optimizer
efficientnet-v2-m	416/5	35	3e-4	adamW

A5 Training Method(s)

Trainning details for classification part

ImageNet pre-trained models were used for backbone. Binary cross entropy loss as loss function for classification branch and segmentation branch. Some other model training details are shown belows:

1. All models (except efficientnet-v2-m) were pretrained with 256 size, efficientnet-v2-m were pretrained with 288 size.
2. Total 35 epochs of training were carried for each model, The learning rate for the first 5 epochs is 5e-4. From 5 epoch to 35 epoch adopts cyclic learning rate, every 5 epochs, the learning rate drops from 5e-4 to 5e-5.
3. Used stochastic weight averaging to ensemble each fold's different snapshots.
4. Cutmix and mixup data augmentation were used with a probability of 0.33 per batch.
5. Used mask averaging to ensemble different models, the weights of each model are equal.
6. Stochastic weight averaging (SWA) improve a lot in cross validation and public leaderboard score.

Trainning details for segmentation part

Classification models were used as pre-trained models. Binary cross entropy loss as loss function for classification branch and combined loss function were used for segmentation branch. Some other model training details are shown belows:

1. All models (except efficientnet-v2-m) were used classification model(trained with 320 size) as pretrained model, efficientnet-v2-m were used classification model(trained with 352 size) as pretrained model.
2. Total 35 epochs of training were carried for each model, The learning rate for the first 5 epochs is 3e-4. From 5 epoch to 35 epoch adopts cyclic learning rate, every 5 epochs, the learning rate drops from 3e-4 to 3e-5.
3. Some other details are same as classification models setting.

Important points for solution

There are mainly five points that are important for our solution, which are listed as following.

1. Using the object detection model to crop key regions for classification and segmentation, which reduces my GPU resource consumption a lot.

2. Using a separate segmentation model for segmentation, because there is a lot of noise in the original labeled data.
3. Using a lot of "heavy" models, in this competition, the "heavy" model performed better.
4. Both cutmix and mixup helped me with classification and segmentation.
5. Stacking 5 slice images as input is better than stacking 3 slice and single slice.

A7 Simple Features and Methods

Although the performance can be improved by ensemble, but the effect of single model and single fold is actually good enough. By adding the detection model and the single fold Unet-efficientnet-v2-l classification model, I can get private/public=0.87211/0.88224, and by adding a single fold segmentation model, I can get private/public 0.87723/0.88451.

A8 Model execution time

All models are trained with a A5000 GPU.

- classification models

All classification models takes about 4 ~ 5 days to train a 5 folds model (including 256-scale pre-training). Most models take about 10 to 20 minutes to train for 1 epoch.

- segmentation models

The efficientnet-v2-l(v1) is use classification models(size=320) as prtrained model, The efficientnet-v2-l(v2) is use segmentation models(size=384) as prtrained model. All segmentation models takes about 3 ~ 4 days to train all 5 folds models (using the classification model or segmentation model as pre-training).

A9 Reference

[1] Ronneberger O, Fischer P, Brox T. U-net: Convolutional networks for biomedical image segmentation[C] International Conference on Medical image computing and computer-assisted intervention. Springer, Cham, 2015: 234-241.
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[6] Huang G, Li Y, Pleiss G, et al. Snapshot ensembles: Train 1, get m for free[J]. arXiv preprint arXiv:1704.00109, 2017.